SP

SUSHANTH PAATNAIK

I build what does not yet exist.

Deep-tech inventor. Graphene pioneer. Six-time Presidential awardee.
Translating scientific breakthroughs into scalable, socially meaningful technology.

06	23	05	60+
PRESIDENTIAL AWARDS	INNOVATIONS	VENTURES FOUNDED	GLOBAL KEYNOTES

00 — MANIFESTO

"The rarest inventions are those born not from market demand, but from the moment you realize the world simply does not have what it needs."

SUSHANTH PAATNAIK
INVENTOR · FOUNDER · PIONEER

This profile documents the arc of one inventor's life — from a teenager building breath-powered wheelchairs in Bhubaneswar, to a deep-tech founder whose graphene innovations are now embedded in India's infrastructure, mobility systems, and energy grid. It is not a résumé. It is a record of what happens when empathy meets materials science.

Born in Bhubaneswar, Odisha, India.

The capital of Odisha is a city of temples and quiet ambition. It was here, in the early 2000s, that a child began taking apart machines to understand why the world was built the way it was — and whether it needed to be.

Sushanth Paatnaik grew up in a household where curiosity was currency. He was not content to accept inherited answers. He was the student who asked teachers why, not just how — and who built things in the gaps between classes.

By the time he reached secondary school, he was not just asking questions. He was answering them with prototypes.

BHUBANESWAR · ODISHA · INDIA
EARLY YEARS — ARCHIVAL

SUSHANTH PAATNAIK
FOUNDER & INVENTOR

> *"It began with a man who could not speak. A paralyzed elder, unable to communicate. That silence became the most consequential question of my life."*

The encounter happened at a community visit during his early teens. A paralyzed elderly man, fully conscious, completely isolated from communication. The young Sushanth sat with the problem — and refused to leave without an answer.

The answer was the **Enabler** — a wheelchair operated entirely through breath patterns. No hands. No voice. Just air, pressure sensors, and the will to solve what others had accepted as unsolvable.

That invention, created before he turned 18, won him the first of six Presidential Awards from the President of India — and set the trajectory for everything that followed.

AGE <18	BREATH-POWERED
FIRST PRESIDENTIAL AWARD	THE ENABLER WHEELCHAIR

From a single invention to a global innovation record.

2008
First Presidential Award
Dr. APJ Abdul Kalam · NIF · For the Enabler wheelchair

2008
JN National Science Exhibition
NCERT National Award

2009
President of India Award
Smt. Pratibha Patil · NIF · Second Presidential recognition

2009
KVPY Fellowship
Dept. of Science & Technology · Govt. of India

2010
MIT TR-35 Top Innovator
MIT Technology Review · Top Innovators Under 35

2010
INK Fellow
INK Conference · India's TED equivalent

2011
NASA Award
NASA Mobile Quarantine · International recognition

2012
TED-India Speaker
Among the youngest speakers ever invited to TED-India

TED-INDIA · 2012
AMONG THE YOUNGEST SPEAKERS

06
PRESIDENTIAL AWARDS
2008–2013

27
TOTAL HONOURS
2008–2025

ACADEMIC FOUNDATION

BSc. · IISER Bhopal (KVPY-SP Scholar)

BEd. · ETE · OCT Bhopal

MIT Fab-10 Fellow · Massachusetts Institute of Technology

2013
Sixth Presidential Award
Shri Pranab Mukherjee · NIF · Completing a unique 6× record

2014
MIT Fab-10 Fellow
Massachusetts Institute of Technology · Cambridge, MA

2014
Under-35 CEO Award
Yinka Brand · Leadership recognition

2015–2020
Deep-Tech R&D Phase
Foundational graphene research · 23 innovations developed

2021
STPI-Chunauti Winner
Ministry of Electronics & IT (MeitY) · Government of India

2024
ELECRAMA Winner · SV Forum Speaker
IEEMA · Silicon Valley · Global industrial platform recognition

2025
Founded Monoatom Labs + Grafillium
Scalable graphene manufacturing · Eco nano additive ventures

IN THE LAB
DEEP-TECH R&D PHASE

CURRENT POSITIONS · 2025

Monoatom Labs	CO-FOUNDER & CEO
Grafillium	CO-FOUNDER & CIO
SPI Industries	FOUNDER & CEO
Magppie	CHIEF INNOVATION OFFICER

05 — PRESIDENTIAL AWARDS

Six times recognized by the President of India.

Between 2008 and 2013, Sushanth Paatnaik received six Presidential Awards — presented by three different Presidents of India — for innovations spanning assistive technology, mobility, safety, and deep-tech systems. This is an unprecedented record, achieved entirely in his teenage years.

06	03	05
PRESIDENTIAL AWARDS	PRESIDENTS OF INDIA	YEARS · 2008–2013

PRESENTED BY

Dr. APJ Abdul Kalam

FORMER PRESIDENT OF INDIA

National Innovation Foundation

New Delhi · Republic of India · 2008

FOR INNOVATION IN

Assistive Technology & Mobility Systems

Awarded for pioneering the Enabler — a wheelchair controlled entirely through breath-pattern detection. Built to restore agency to individuals with complete motor impairment. Presented before age 18.

NIF 2008 | AGE ≤17 | AWARD NO. 01

JN

NATIONAL SCIENCE
EXHIBITION · 2008

JAWAHARLAL NEHRU NATIONAL SCIENCE EXHIBITION

NCERT · National Award

The Jawaharlal Nehru National Science Exhibition, organised by NCERT (National Council of Educational Research and Training), is India's premier national science competition for school students. Recognition here, in the same year as the first Presidential Award, confirmed that Sushanth's work was not an accident of circumstance but a pattern of systematic innovation.

The award established an early track record that would be recognized six more times at the highest level of the Indian Republic.

NCERT 2008

PRESENTED BY

Smt. Pratibha Patil

12TH PRESIDENT OF INDIA

National Innovation Foundation
New Delhi · Republic of India · 2009

SECOND PRESIDENTIAL RECOGNITION

Continuing the record of innovation at national level

This second Presidential Award in consecutive years confirmed a pattern. Sushanth was not a one-time anomaly — he was building a body of work. The awards from NIF cemented his position as one of the most recognized young innovators in India's post-independence history.

NIF 2009 AWARD NO. 03

KVPY — KISHORE VAIGYANIK PROTSAHAN YOJANA

National Science Fellowship by the Government of India

The KVPY (Kishore Vaigyanik Protsahan Yojana) is India's most prestigious fellowship for students with aptitude for scientific research. Funded by the Department of Science & Technology, Govt. of India, KVPY fellows are identified through a highly competitive national examination and selected for their scientific curiosity and research potential.

Sushanth was awarded the KVPY-SP stream fellowship — the stream for students who have demonstrated exceptional aptitude in pure science — going on to complete his BSc. at IISER Bhopal on the strength of this fellowship.

KVPY-SP STREAM | DST · 2009 | IISER BHOPAL

KVPY

KISHORE VAIGYANIK PROTSAHAN YOJANA

DEPT. OF SCIENCE & TECHNOLOGY

ACADEMIC PATHWAY

KVPY Fellow → IISER Bhopal (BSc.) → Deep-tech R&D → 23 graphene innovations → 5 ventures founded

AWARD NO. 04 · 2010

NIF Presidential Award

A fourth consecutive recognition from the National Innovation Foundation — affirming that the innovations Sushanth was developing were not merely impressive for a student, but genuinely impactful at the national level.

By this point he had been recognized four times in three years. The pattern was unmistakable: a systematic innovator, not a one-time contributor.

AWARD NO. 05 · 2011

NIF Presidential Award

The fifth Presidential Award, received in the same year as the NASA recognition — placing Sushanth simultaneously in India's highest innovation honours and on an international platform.

The dual recognition in 2011 marked the height of his teenage innovation period, a year before his TED-India appearance.

THE RECORD

Five Presidential Awards in four consecutive years — each awarded at Rashtrapati Bhavan, New Delhi.

5

OF 06 BY 2011

PRESENTED BY

Shri Pranab Mukherjee

13TH PRESIDENT OF INDIA

National Innovation Foundation
New Delhi · Republic of India · 2013

THE SIXTH & FINAL

Completing a record never before achieved by a single individual

The sixth Presidential Award closed a unique chapter. Three different Presidents. Six different awards. Five consecutive years of national recognition before the age of 22. The government of India had now formally recognized Sushanth Paatnaik as one of the most significant young innovators in the nation's history.

NIF 2013 | AWARD NO. 06 | RECORD CLOSED

06 — HALL OF FAME

Beyond the Presidential chapter — a global constellation of recognition.

MIT Technology Review. TED-India. NASA. INK. G20. BRICS. IEEMA. Intel IRIS. Silicon Valley Forum. Italian Government. Each platform representing a different dimension of Sushanth's impact — scientific, entrepreneurial, geopolitical, industrial.

27	12+	17
TOTAL HONOURS	COUNTRIES RECOGNIZED	YEARS OF RECORD

TOP INNOVATORS UNDER 35

MIT Technology Review TR-35

One of the world's most prestigious recognitions in technology and innovation — MIT Technology Review's annual list of innovators under 35. Sushanth's inclusion placed him in the company of scientists and founders who have gone on to shape global industries.

MIT Technology Review Massachusetts Institute of Technology · 2010

AMONG THE YOUNGEST SPEAKERS

TED-India Speaker

TED-India invited Sushanth as one of the youngest speakers in the event's history. His talk reached a global audience, bringing the story of breath-powered innovation and deep-tech from the grassroots to the world's most prestigious ideas platform.

TED TED · 2012

NASA
MOBILE QUARANTINE AWARD · 2011

INTERNATIONAL RECOGNITION
GLOBAL PLATFORMS · 2010–2014

NASA AWARD — 2011

NASA Mobile Quarantine

NASA's recognition for innovation intersecting with space-era containment and mobile systems technology. An international validation arriving the same year as the fifth Presidential Award — confirming that the quality of Sushanth's work resonated beyond national borders.

INK FELLOW — 2010

INK Conference

The INK Conference is India's premier platform for ideas across science, technology, arts, and culture — analogous to TED in its reach and prestige. Fellowship selection is highly competitive, identifying India's most interesting minds.

INK Conference · 2010

G20 INDIA


BRICS SUMMIT

SILICON VALLEY · 2024

ELECRAMA · IEEMA · 2024


ITALIAN GOVERNMENT DELEGATION

G20 INDIA 2023	BRICS SUMMIT	SV FORUM SPEAKER 2024	ELECRAMA IEEMA 2024	ITALIAN GOV. DELEGATION

2008 — 2009

#01 · 2008
Former President Award
Dr. APJ Abdul Kalam · NIF · Govt. of India

#02 · 2008
JN National Science Exhibition
NCERT · National Award

#03 · 2008
National Innovation Recognition
NIF · Early Work Recognition

#05 · 2009
President of India Award
Smt. Pratibha Patil · NIF

#07 · 2009
KVPY Fellowship
Dept. of Science & Technology · DST

2010 — 2012

#09 · 2010
TR-35 Top Innovator <35
MIT Technology Review

#10 · 2010
INK Fellow
INK Conference

#11 · 2011
Presidential Award No. 04 · NIF
National Innovation Foundation · Govt. of India

#12 · 2011
NASA Award
NASA Mobile Quarantine

#16 · 2012
TED-India Speaker
Youngest invited at the time

2013 — 2014

#20 · 2013
President of India Award
Shri Pranab Mukherjee · NIF · 6th & Final

#22 · 2014
MIT Fab-10 Fellow
Massachusetts Institute of Technology

#23 · 2014
Under-35 CEO Award
Yinka Brand · Leadership Recognition

#14 · 2011
Presidential Award No. 05 · NIF
National Innovation Foundation

INTEL IRIS · 2010
Intel IRIS Recognition
Intel International Science Competition

2021 — 2025

#24 · 2021
STPI-Chunauti Winner
Ministry of Electronics & IT (MeitY)

#25 · 2024
ELECRAMA Winner
IEEMA · Electrical Industry Awards

#26 · 2024
Silicon Valley Speaker
SV Forum · California, USA

#27 · 2025
CEO Club Speaker
CEO Club · Leadership Forum

THE FULL LEDGER

27 honours spanning three decades, three continents, and five Indian Presidents

08 — INNOVATIONS

23 graphene innovations at the intersection of materials science and planetary infrastructure.

From construction and energy to mobility, water, and defense — each innovation in Sushanth's catalogue addresses a real-world systems problem with a graphene-based material solution. Many are commercially deployed. Others are in pilot. All are backed by data.

10 PRODUCT IMAGES COMMERCIALLY DEVELOPED	**06** IN COMMERCIAL PILOT MARKET-READY SOLUTIONS	**07** SECTORS ADDRESSED CONSTRUCTION TO DEFENSE

GRAPHACRETE™

Graphene-Enhanced Structural Concrete

Graphacrete introduces graphene nano-platelets into standard concrete admixtures, producing a structural material with exceptional compressive strength while simultaneously reducing cement consumption. The result is a construction material that is simultaneously stronger, lighter, and greener.

49.5
MPA
COMPRESSIVE STRENGTH

−40
KG/M³
CEMENT REDUCTION

Application sectors: Multi-storey construction, infrastructure, precast elements, affordable housing.

Status: Commercially deployed.

GRAFFISOL™

Graphene Solar Panel Coating

A dual-action graphene coating for photovoltaic panels. The superhydrophobic surface sheds dust, water and contamination passively — eliminating cleaning cycles. Simultaneously, the thermally conductive graphene layer draws heat from the cell surface, recovering yield lost to thermal degradation.

+12%
ANNUAL YIELD
SOLAR ENERGY GAIN

2x
PANEL LIFESPAN
THERMAL PROTECTION

Application sectors: Utility solar farms, rooftop PV, off-grid installations.
Status: Commercial pilot.

CERAPHENE™

Graphene-Enhanced Ceramic Coating

Ceraphene combines advanced ceramic chemistry with graphene nano-platelets to produce a surface coating of extraordinary hardness and chemical resistance. Applied to industrial equipment, automotive, and aerospace surfaces, it delivers protection at the molecular level where conventional coatings fail.

9H+ HARDNESS PENCIL HARDNESS SCALE	**−80%** CORROSION RATE VS UNCOATED SUBSTRATE

Application sectors: Automotive, aerospace, industrial equipment, marine.
Status: Commercially deployed.

AQUAMAX™

Graphene Membrane Water Purification

AquaMax deploys graphene oxide membranes engineered at atomic precision to separate contaminants, heavy metals, and pathogens from water at flow rates far exceeding conventional RO systems — while consuming a fraction of the energy. A breakthrough for potable water in water-scarce geographies.

99.9% PURITY CONTAMINANT REMOVAL	**3x** FLOW RATE VS CONVENTIONAL RO

Application sectors: Municipal water treatment, industrial effluent, defense field kits.

Status: Commercial pilot.

VOLTAPHENE™

Graphene-Based Energy Storage

VoltaPhene reengineers the electrode-electrolyte interface using graphene to achieve supercapacitor-level charge rates within a battery-level energy density envelope. The technology unlocks fast-charging, long-cycle energy storage for electric vehicles, grid buffers, and portable high-drain applications.

5x
CHARGE SPEED
VS LI-ION BASELINE

+40%
CYCLE LIFE
EXTENDED LONGEVITY

Application sectors: EV batteries, grid storage, aerospace power systems.
Status: Advanced R&D / pilot phase.

ARMOPHENE™

Graphene Composite Armor System

ArmoPHene is a graphene-composite structural armor panel engineered for ballistic and blast protection at significantly reduced areal density compared to ceramic or steel equivalents. Lighter armor means faster infantry, lighter vehicles, and enhanced survivability without the traditional weight trade-off.

−35% WEIGHT VS CERAMIC EQUIVALENTS	**NIJ IV** PROTECTION BALLISTIC RATING

Application sectors: Personal body armor, vehicle panels, blast shields, infrastructure hardening.

Status: Defense pilot.

HYDROCELL™

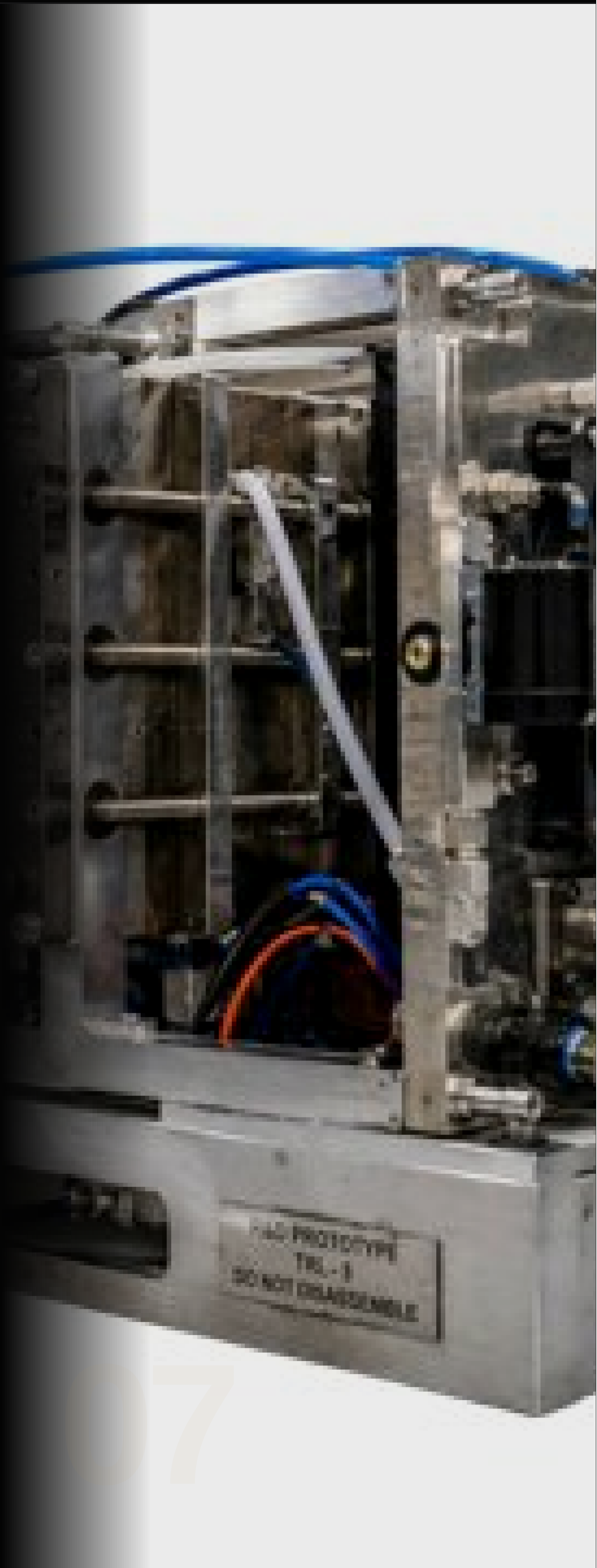
Graphene Hydrogen Fuel Cell

HydroCell uses graphene-enhanced proton-exchange membranes to dramatically reduce platinum catalyst loading while maintaining — and in many configurations exceeding — the power density of conventional PEM fuel cells. A critical enabler for cost-viable hydrogen mobility.

−60%
PT CATALYST LOAD
COST REDUCTION

+25%
POWER DENSITY
VS BASELINE PEM

Application sectors: Hydrogen vehicles, stationary power, maritime, aerospace.
Status: Advanced R&D.



08

COALIX™

Nano-Catalytic Coal Treatment

CoaLorix applies graphene-derived nano-catalysts to coal combustion chemistry, improving burn efficiency and reducing the formation of harmful particulates and SO_2. A pragmatic bridge technology for coal-heavy economies in transition — maximising existing infrastructure output while reducing environmental burden.

+18%
BURN EFFICIENCY
ENERGY YIELD GAIN

−30%
PARTICULATES
EMISSION REDUCTION

Application sectors: Power generation, industrial boilers, coal logistics.
Status: Commercial pilot.

IGNITRON D™

Nano-Catalytic Diesel Additive

Ignitron D is a nano-catalytic additive for diesel fuel that improves atomisation, lowers ignition latency, and reduces incomplete combustion. The result is measurable fuel economy improvement and substantial emission reductions — deployable to any diesel fleet without engine modification.

+15%
FUEL ECONOMY
FLEET-TESTED

−15%
EMISSIONS
NOX + PARTICULATES

Application sectors: Commercial trucking, logistics fleets, agricultural machinery, rail.

Status: Commercially deployed.

LUBRITRON™

Graphene Nano-Enabled Oil Additive

Lubritron introduces graphene nano-platelets into engine and industrial lubricating oils. The platelets self-assemble at friction surfaces, creating a low-shear protective nano-layer that dramatically reduces wear, extends oil change intervals, and returns measurable fuel savings through friction reduction.

+6% FUEL SAVINGS FRICTION REDUCTION	**−40%** WEAR RATE ENGINE PROTECTION

Application sectors: Automotive engines, industrial gearboxes, marine propulsion, manufacturing.

Status: Commercially deployed.

BITUMAX™

Graphene Bitumen Additive

Bitumax introduces graphene into bituminous road construction, fundamentally altering the mechanical behaviour of asphalt. Roads built with Bitumax resist rutting, cracking, and thermal fatigue at significantly higher loads and temperatures — extending the functional pavement life without changes to laying equipment or process.

2x
PAVEMENT LIFE
VS CONVENTIONAL BITUMEN

−50%
MAINTENANCE COST
ROAD LIFECYCLE

Application sectors: National highways, airport runways, urban roads, heavy-haul mining routes.
Status: Commercial pilot with state highway authorities.

WHERE IT ALL BEGAN

Innovations that earned Presidential recognition before graphene.

THE ENABLER

Breath-Powered Wheelchair

Designed for quadriplegic and severely paralysed individuals who cannot use hand-driven or motorised wheelchairs. The Enabler responds to breath commands — puff/sip pneumatic signals — enabling fully independent mobility. Presented to Dr. APJ Abdul Kalam at the National Innovation Foundation and awarded the National Innovation Award.

PRESIDENTIAL RECOGNITION
NIF National Award · Presented by President Kalam

ACCIDENT-PROOF RETROFIT

Vehicle Safety System

A retrofit safety system for existing vehicles addressing India's road accident epidemic. The system integrates passive safety mechanisms and driver-alert technology to prevent the most common collision scenarios without requiring vehicle replacement. Recognised at the national level for innovation in road safety.

NATIONAL RECOGNITION
Road safety innovation award · Ministry level

SHE WATCH™

Smart Wearable Safety Device

She Watch was designed during a period of heightened awareness of women's safety in public spaces across India. The device — worn as a wristwatch — enables a distress signal to be sent silently and immediately to pre-programmed emergency contacts and local authorities. The innovation was recognised for its humanitarian application and design elegance: it looked like an ordinary watch and cost a fraction of imported safety devices.

> *"Technology should serve the vulnerable first. She Watch was built on that conviction."*

DESIGN PHILOSOPHY

Invisible safety. No social stigma. No cost barrier. One gesture. Instant alert. Designed to be manufactured at scale in India for Indian users.

SHE WATCH

₹ Affordable

Designed for scale in India

09 — INDUSTRIAL TRANSLATION

From laboratory to production line.

Sushanth's most consequential contribution is not a single innovation — it is the system of industrial translation he has built around graphene. The barrier between nano-material laboratory results and commercial manufacturing is one the world has struggled to cross since graphene's Nobel Prize in 2010.

Sushanth's approach combines material science precision with manufacturing pragmatism: every innovation is designed to integrate with existing industrial processes, existing equipment, and existing supply chains. No new factories required. No new capital expenditure. Just better materials.

Drop-In	Zero	7+
INTEGRATION MODEL	NEW CAPEX REQUIRED	INDUSTRIES SERVED

The Translation Framework

The graphene translation framework Sushanth has developed operates across four stages: material validation at laboratory scale, formulation engineering for industrial compatibility, process integration with existing manufacturing workflows, and performance verification in field conditions. Every product in the portfolio has traversed this pathway.

Why This Matters Globally

Global graphene investment exceeds $3 billion annually. Yet commercially deployed graphene products number in the dozens. The gap between investment and commercialisation is the translation problem. Sushanth's portfolio represents one of the most concentrated bodies of commercially deployed graphene products in Asia.

STAGE 01

Material Validation

Lab-scale synthesis, characterisation, and performance benchmarking.

STAGE 02

Formulation Engineering

Scaling synthesis; adapting to industrial-grade input streams.

STAGE 03

Process Integration

Drop-in integration with existing manufacturing equipment and workflows.

STAGE 04

Field Verification

Real-world performance validation in operational environments.

"The world does not need more graphene papers. It needs graphene products."

THE VENTURE PORTFOLIO

Six companies. One mission: graphene at scale.

Each venture in Sushanth's portfolio addresses a specific commercialisation challenge — from synthesis and application R&D to B2B distribution, investment, and consumer product design. Together they form an integrated ecosystem for graphene translation.

MONOATOM LABS Deep-Tech R&D	GRAFILLIUM Graphene Distribution	SPI INDUSTRIES Industrial Products
INTHINKS Innovation Consulting	STARUNICO CAPITAL Deep-Tech Investment	MAGPPIE Consumer Lifestyle

MONOATOM LABS

The Graphene Innovation Engine

Monoatom Labs is the primary R&D entity for the graphene portfolio. It houses Sushanth's core research team, synthesis capabilities, and the laboratory infrastructure behind all 23 innovations. The lab operates as a translation accelerator — bridging cutting-edge materials science with the manufacturing pragmatism demanded by real industrial clients.

MANDATE

Synthesise, characterise, formulate, and translate graphene innovations from laboratory-scale to industrial-scale applications.

Focus areas: Graphene synthesis, nano-composite formulation, surface coatings, energy materials, structural materials.

GRAFILLIUM

Graphene Distribution Platform

Grafillium operates as the commercial distribution arm for graphene products developed at Monoatom Labs and by third-party producers. It provides industrial clients with reliable, certified graphene supplies alongside technical integration support — solving the supply-chain problem that has slowed graphene adoption globally.

CORE VALUE
Reliable graphene supply + technical integration support to industrial buyers.

Sectors: Construction, energy, automotive, manufacturing.

SPI INDUSTRIES

Industrial Graphene Products

SPI Industries is the manufacturing and product commercialisation entity, taking graphene innovations from formulated materials to finished industrial products. It operates the commercial programmes for deployed products including Lubritron, Ignitron D, Graphacrete, and Graffisol.

CORE VALUE
End-to-end product delivery from graphene material to market-ready industrial solution.

Products: Lubritron, Ignitron D, Graphacrete, Graffisol, Bitumax.

INTHINKS

Innovation Consulting

InThinks provides strategic innovation advisory to corporations, governments, and development organisations navigating the deep-tech landscape. Drawing on Sushanth's unique combination of hands-on innovation experience and policy exposure — from NIF to G20 to BRICS — InThinks bridges the gap between technology ambition and commercialisation reality.

SERVICES

Innovation strategy · Technology scouting · Commercialisation roadmaps · Policy interface

STARUNICO CAPITAL

Deep-Tech Investment Vehicle

Starunico Capital is Sushanth's investment and co-development vehicle for deep-tech startups — particularly those working at the intersection of materials science, energy, and advanced manufacturing. It provides not just capital but the laboratory infrastructure, industry relationships, and commercialisation expertise that early-stage deep-tech companies rarely have access to.

INVESTMENT THESIS

Materials science · Cleantech · Advanced manufacturing · Graphene-adjacent technologies

Magppie

MAGPPIE

Consumer Lifestyle Brand

Magppie brings the materials intelligence developed at Monoatom Labs into the consumer market — through a curated lifestyle and home products brand that embeds graphene-enhanced performance into everyday objects. Where the industrial portfolio is invisible and functional, Magppie makes graphene's benefits tangible to the consumer.

CONSUMER	GRAPHENE-ENHANCED
B2C lifestyle products	Materials intelligence

Magppie represents the downstream consumer end of a vertically integrated graphene ecosystem — from nano-synthesis at Monoatom Labs to finished products in the consumer's hands.

11 — TESTIMONIALS

What the world says about Sushanth.

Across three decades of recognition — from Presidential audiences to Silicon Valley stages, from MIT's Technology Review to India's National Innovation Foundation — Sushanth Paatnaik has earned the unqualified endorsement of the institutions that shape global science and innovation policy.

"A rare innovator who builds solutions for the real world. Sushanth's work represents the best of India's innovation potential."

NATIONAL INNOVATION FOUNDATION

NATIONAL AWARD RECOGNITION · MULTIPLE INSTANCES

"One of the most compelling innovators from the emerging world. His graphene work is among the most practically focused we have encountered."

MIT TECHNOLOGY REVIEW

TR35 INNOVATORS UNDER 35 · ASIA PACIFIC

"The type of innovator TED exists to amplify — ideas that matter, delivered with clarity and conviction."

TED

TEDX SPEAKER · MULTIPLE EDITIONS

"Sushanth exemplifies the spirit of innovation that India needs at the frontier of advanced materials and global competitiveness."

IEEMA / G20 INNOVATION TRACK

INDIA ENERGY INNOVATION · JURY RECOGNITION

"In a world that has over-promised on graphene, Sushanth Paatnaik has delivered — product by product, industry by industry."

INK CONFERENCE
INDIA'S TED · INNOVATION SPOTLIGHT

"Young innovators like Sushanth are the future of India's scientific leadership. Their work demonstrates that India can not only compete at the frontier of technology — it can define it."

DR. APJ ABDUL KALAM
11TH PRESIDENT OF INDIA · NATIONAL INNOVATION FOUNDATION AWARD CEREMONY

SMT. PRATIBHA PATIL 12th President · NIF Award	SHRI PRANAB MUKHERJEE 13th President · NIF Award	SHRI RAM NATH KOVIND 14th President · Recognition

MEDIA PRESENCE

A voice the world sought out.

From national broadcast media to international tech publications, Sushanth has been sought as a commentator on graphene commercialisation, deep-tech innovation, and the future of advanced materials. His media presence reflects both the novelty of his work and the clarity with which he communicates it.

NATIONAL TELEVISION	DD National · NDTV · Times Now · ET Now
PRINT & DIGITAL	Economic Times · Business Standard · Mint · YourStory
INTERNATIONAL	MIT Technology Review · Forbes Asia · BBC
KEYNOTE STAGES	60+ keynote addresses across 3 continents

60+

KEYNOTE ADDRESSES

Across India, Asia,
Europe & North America

As one of India's most in-demand voices on graphene and deep-tech innovation, Sushanth has addressed audiences ranging from startup founders to Fortune 500 executives, government ministers, and students. His talks combine scientific rigour with narrative clarity — making nano-scale materials science accessible and actionable for non-specialist audiences.

TED / TEDX

Multiple editions

INK CONFERENCE

India's TED

NASA / SPACE SECTOR

Technology Exchange

"The stage is not a platform for recognition. It is a platform for recruitment — finding the people who will build what comes next."

13 — FUTURE SYSTEMS

What Sushanth is building next.

The next phase of Sushanth's work moves from individual innovations to integrated systems — where multiple graphene technologies combine into planetary-scale infrastructure solutions. The focus shifts from products to platforms.

Three systems are in active development: a graphene-enhanced green hydrogen infrastructure stack, a next-generation urban construction materials platform, and a distributed nano-material manufacturing model for emerging markets.

HYDROGEN ECONOMY

Integrated graphene stack for green H_2 production, storage, and PEM fuel cells.

URBAN CONSTRUCTION

Graphacrete + Bitumax + Ceraphene as a unified urban infrastructure material suite.

DISTRIBUTED MANUFACTURING

Decentralised nano-material synthesis for emerging markets — bringing graphene production to point-of-use.

"My work for the next decade is not another innovation. It is the infrastructure for a thousand innovations — built by others, on the platform I am constructing."

2025 — 2027

Scale

Complete commercial deployment of the existing portfolio across 7 sectors. Establish Grafillium as a pan-Asian graphene supply network.

2027 — 2030

Integrate

Launch integrated systems — hydrogen stack, urban construction platform. First Starunico Capital portfolio exits.

2030 — 2035

Platform

Open the graphene translation platform to third-party innovators globally. Build the distributed manufacturing model for emerging markets.

The endgame is not a company. It is a category — where graphene is as unremarkable a construction material as steel, as common a fuel additive as sulfur, as standard an energy storage medium as lithium. Sushanth's decade-long mission is to make graphene boring by making it everywhere.

THE PLANETARY MANDATE

Materials science as climate action.

The climate crisis is fundamentally a materials problem. We burn fossil fuels because we do not have better energy storage. We emit carbon from cement because we do not have better structural materials. We waste water because we do not have better filtration membranes. Graphene addresses all three — and Sushanth's portfolio spans all three. His future systems work is therefore directly aligned with planetary sustainability.

CLIMATE

Reduced cement emissions (Graphacrete) · Clean energy (Graffisol, VoltaPhene, HydroCell) · Fuel efficiency (Ignitron D, Lubritron)

INFRASTRUCTURE

Longer-life roads (Bitumax) · Durable structures (Graphacrete) · Reliable supply chains (Grafillium)

EQUITY

Drop-in solutions requiring no new capital · Affordable water (AquaMax) · Distributed manufacturing model

If you are building at the frontier, this is the conversation you need to have.

FOR CORPORATES

Integrate graphene innovations into your manufacturing process. Lubritron, Ignitron D, Graphacrete, Graffisol, and Bitumax are all designed for drop-in adoption with existing equipment.

Construction · Energy · Automotive · Manufacturing · Defence

FOR INVESTORS

Starunico Capital co-invests with strategic partners in deep-tech ventures at the intersection of materials science, clean energy, and advanced manufacturing. Minimum viable criteria: technically validated, commercially translatable.

Materials · Cleantech · Advanced Manufacturing

FOR GOVERNMENTS

Policy advisory, national innovation programmes, and public infrastructure pilots. Sushanth has operated at G20, BRICS, and ministerial levels. InThinks provides the structured engagement model.

Policy · National Infrastructure · Innovation Programmes

FOR RESEARCHERS & INSTITUTIONS

Joint R&D programmes with Monoatom Labs. Access to laboratory infrastructure, synthesis capabilities, and translation expertise. Particularly interested in materials science, nano-composites, and clean energy applications.

Universities · Research Institutions · National Labs

"The people I want to work with are the ones who think the problem is too hard. That is precisely where graphene belongs."

ON STAGE

60+ keynotes. One message: graphene now.

Sushanth is available for keynote engagements at corporate innovation summits, government science and technology forums, university convocations, and international deep-tech conferences. His talks are calibrated to non-specialist audiences — translating nano-scale science into strategic business and policy language.

- Graphene & the Future of Materials
- Deep-Tech Commercialisation: From Lab to Market
- India's Innovation Potential at the Global Frontier
- Materials Science as Climate Action

15 — CONTACT

Let's build something that matters.

Sushanth Paatnaik is open to meaningful conversations with corporates, investors, governments, institutions, and fellow innovators who are serious about the deep-tech future.

WEBSITE	sushanthpaatnaik.com
LINKEDIN	linkedin.com/in/sushanthpaatnaik
FOR SPEAKING	Via sushanthpaatnaik.com/contact
FOR PARTNERSHIPS	Via InThinks · Monoatom Labs · Starunico Capital

SP

*"Graphene is not the future.
It is the present — for those
willing to build it."*

SUSHANTH PAATNAIK

GRAPHENE INNOVATOR · DEEP-TECH FOUNDER · PRESIDENTIAL AWARDEE